Evaluation of Causal Reasoning for Large Language Models in Contextualized Clinical Scenarios of Laboratory Test Interpretation

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Background

- LLMs now often outperform clinicians on medical benchmarks, and research is shifting from answer accuracy to **reasoning quality**.
- Clinicians decide on cause and effect, not association alone, yet **causal reasoning** of LLMs in clinical settings is underexplored.

Objectives

- Evaluate LLM causal reasoning in **laboratory test contexts** across the three levels of Pearl's Ladder of Causation[1]: association, intervention, and counterfactual.

Research Design

- **99 causal questions** generated with Python, 33 per rung.
- **8 blood tests:** HbA1c, creatinine, vitamin D, CRP, cortisol, LDL, HDL, albumin.
- **Causal factors** per test (age, gender, obesity, smoking, physical activity, medication use, inflammation) from MedlinePlus and Mayo Clinic.
- Each question set in a defined clinical context and verbalized as a structured prompt.
- **Models:** GPT-o1 and Llama-3.2-8b-instruct, each giving a definitive answer plus concise reasoning.
- **Four medically trained experts** rated the answer (yes/no) and the reliability of the reasoning (Likert 1–5).

Discussion

- GPT-o1 aligned more closely with expert consensus, particularly on counterfactual and association tasks.
- Interventions reflect widely known clinical relationships, so both models did well.
- Strong answers **traced the causal pathway** and considered confounders such as age, gender, and comorbidities; vague answers scored lower.
- Failures were model-specific: GPT-o1 sometimes gave terse, unexplained answers, while Llama-3.2 was more variable in clarity and alignment with experts.
- Reasoning AUROC was comparable (GPT-o1 0.76 vs. Llama-3.2 0.81); Llama's richer explanations sometimes cost clarity.
- Performance was uneven and sensitive to question framing and clinical nuance.

Conclusions

- 99 lab-based scenarios mapped to Pearl's ladder show **GPT-o1 as the more consistent causal reasoning model**.
- LLMs may be most reliable for well-established interventions and less stable for complex counterfactuals.

Study Pipeline

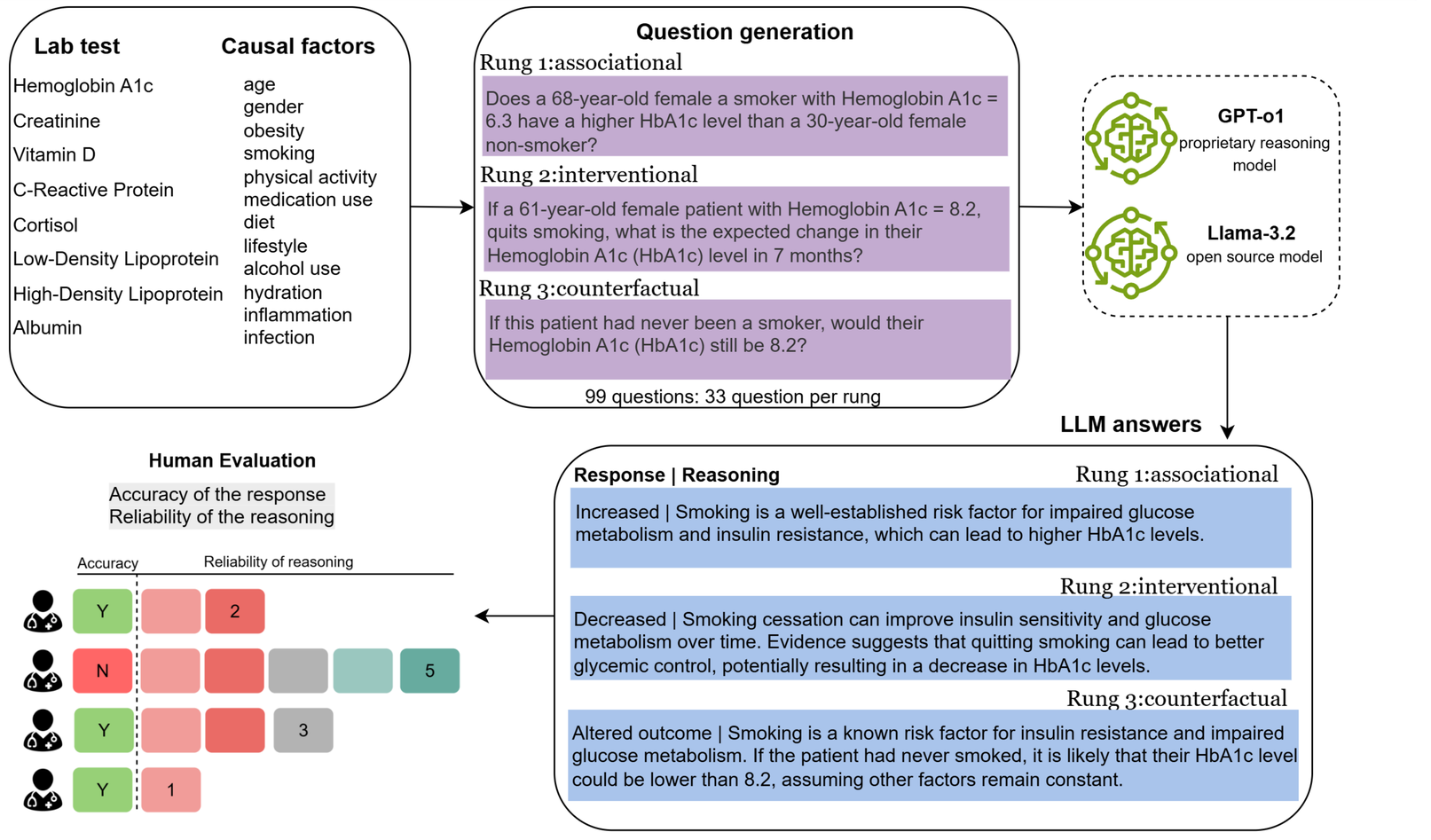
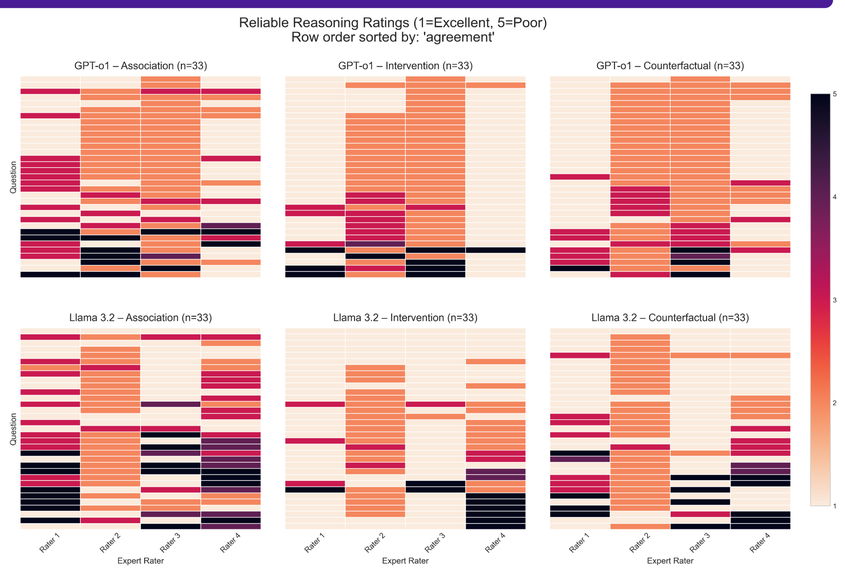
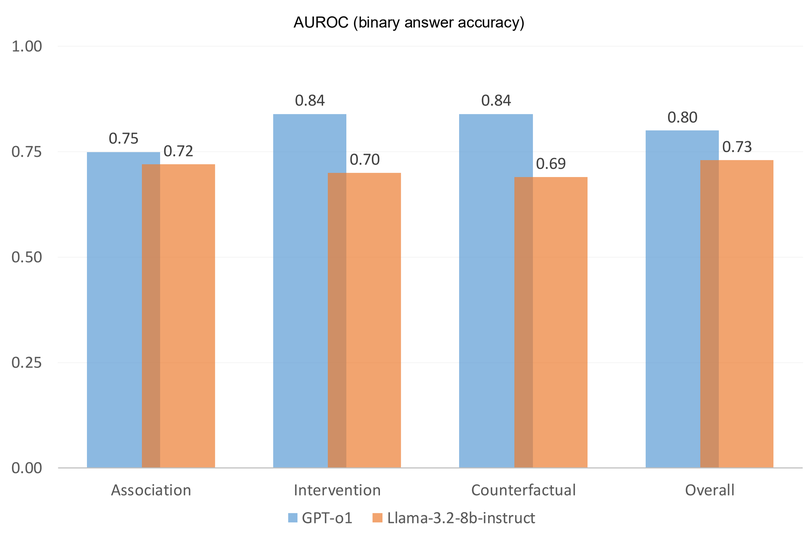


Figure 1. Pipeline for generating and evaluating causal questions and reasoning: lab tests and causal factors generate questions at three rungs, two LLMs answer, and clinicians rate answers and reasoning.

Results

- **GPT-o1 outperformed Llama-3.2 on every rung:** overall AUROC 0.80 ± 0.12 vs. 0.73 ± 0.15.
- GPT-o1 also had higher precision (0.98 vs. 0.92), sensitivity (0.90 vs. 0.84), and specificity (0.70 vs. 0.62), especially on interventions and counterfactuals.
- **Experts agreed most on interventions** (60 of 66 responses judged correct by majority; 0.95 agreement).
- **"Altered outcome" answers** to counterfactuals reached only 0.04 agreement (n = 24).





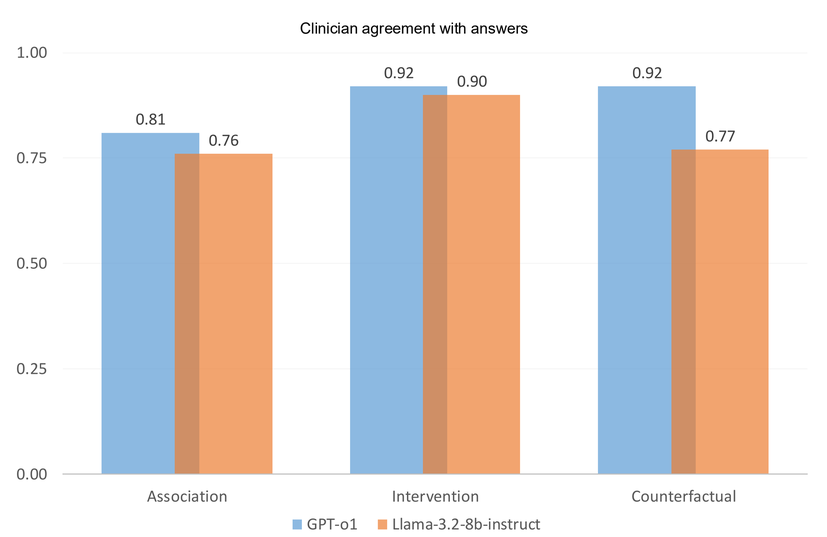


Figure 2. GPT-o1 vs. Llama-3.2-8b-instruct by causal rung: discriminative performance (left) and expert agreement (right).

Future Work

- Expand beyond lab tests to history and physical findings, imaging, genomics, and longitudinal data.
- Add guardrails: retrieval augmentation, explicit assumption disclosure, and uncertainty tagging.

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References

1. Pearl J. An introduction to causal inference. Int J Biostat. 2010;6(2):Article 7.